Integrating Renewables into Nodal Energy Markets

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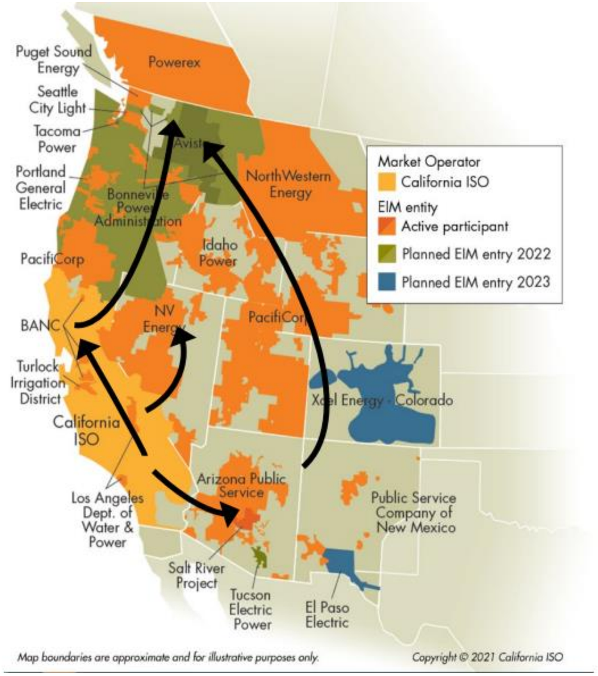
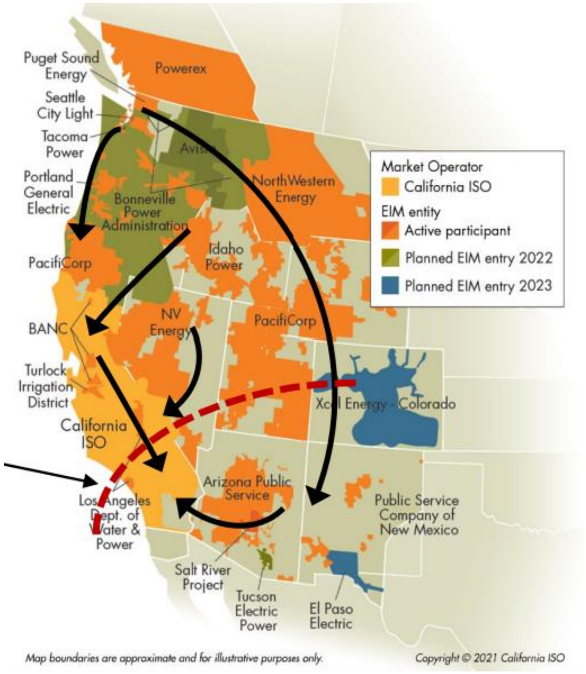
General approaches

1. Forecasting & scheduling
2. Flexible generation resources
3. Energy storage
4. Demand response programs
5. Transmission infrastructure development
6. Advanced grid management systems
7. Integration of Distributed Energy Resources (DERs)
8. Grid modernization

CAISO

- Western Energy Imbalance Market expansion





- Flexible ramping products
  - procure upward & downward ramping capacity over 15- to 30-minute look ahead in the real time market

New Zealand

Demand-side participation

1. **Dispatchable demand**
   - demand-side participants compete with generators to set the spot price
   - for <u>large consumers</u> who buy electricity from the spot market & can modify all / part of their consumption at short notice

2. **Dispatch notification load**
   - allows <u>smaller load stations</u> to directly participate in the spot market
   - for owners of small-scale generation & flexible load (*EV chargers, solar, battery installations*, commercial buildings)

References

1. https://www.caiso.com/Documents/RenewableIntegrationUnlockingDividends.pdf
2. https://www.caiso.com/Documents/EISG-Presentation-Renewable-Integration-in-the-Western-US-Apr17-2023.pdf
3. https://www.transpower.co.nz/system-operator/information-industry/electricity-market-operation/demand-side-participation
4. https://www.westerneim.com/pages/default.aspx
5. https://www.osti.gov/servlets/purl/1801337#:~:text=The%20up%20and%20down%20flexible,ramping%20requirements%20of%20the%20system
6. https://www.transpower.co.nz/system-operator/information-industry/electricity-market-operation/demand-side-participation

Thanks!